



Rounak Singh

Data & Platform Engineer (Scalable Data Pipelines | ETL | Cloud Computing | CI/CD | MLOps)

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Address: Duisburg, North Rhine-Westphalia, 47057

Languages: English (C1) | German (B1)

*** Currently in Germany on valid work permit | Can start immediately | Open to travel and relocation ***

ABOUT ME:

Offering 3+ years of experience building and operating production-grade ETL pipelines on AWS and Snowflake. Proven expertise in Python, AWS Glue, Redshift, and data lake design. Proven ability to process 500K+ daily records, integrate data across multiple systems and improve pipeline reliability and performance for analytics in ML use cases along with CICD automation in multiple enterprise projects.

PROFESSIONAL EXPERIENCE:

Senior Software Engineer (Data Engineering)

Feb 2022 - Feb 2025

Hexaware Technologies Inc.

Atlanta, USA

- Owned and maintained **AWS-based ETL pipelines** (Glue, Redshift, S3, Lambda) processing 500K+ records daily across 40+ tables, enabling reliable data delivery for analytics and **ML** systems for **Fortune 500** Clients.
- Integrated data from 14+ heterogeneous sources, improving accessibility for downstream analytics teams through **APIs**.
- Designed and optimized **data models** and **schema** architectures (star/snowflake) for customer analytics workloads.
- Developed Apache **Airflow** DAGs to orchestrate ETL workflows, handling scheduling, dependencies, **retries**, and monitoring improving system reliability and uptime by 33%.
- Developed **CI/CD** pipelines using **GitLab** and **Terraform** to provision Infrastructure and reduced deployment time by 40%.
- Developed data pipelines on **Databricks** using **PySpark**, managing clusters for high-volume data processing.
- Migrated legacy **enterprise** infrastructure from (Netezza + Talend + Informatica) to (**Snowflake** + **AWS**) to reduce tech debt.

Graduate Research Assistant

June 2020 - Dec 2021

University of Missouri

Missouri, USA

- Designed **reinforcement learning** models (A3C) for optimizing UAV trajectory in dynamic environments.
- Processed **120,000+** geospatial data records for simulation and model evaluation of **last-mile logistics**-based use cases.
- Collaborated with cross-functional University research teams on projects in **robotics** and distributed systems.
- Developed ML pipelines for preprocessing, training, and evaluation using Python, Pandas, Numpy, TensorFlow, and PyTorch.
- Built **feature engineering** pipelines to transform raw datasets into model-ready inputs, improving model performance and reducing data leakage.

EDUCATION:

Masters - Electrical Engineering, **University of Missouri Columbia**, USA

Aug 2019 – Dec 2021

Bachelors - Electronics & Communication Engineering, **GITAM Deemed to be University**, India

Aug 2015 – 2019

SKILLS:

- Data:** Python, SQL, MySQL, PostgreSQL
- Cloud:** AWS (Glue, Redshift, S3, ECS, EMR, Lambda, IAM, SNS, Step Functions), Snowflake, Databricks
- Infra Tools:** Terraform, Git, Github, Bitbucket, Kafka, Airflow, Spark, BigQuery
- CI/CD:** GitLab, Jenkins, Docker, Kubernetes
- Engineering:** Batch & Stream processing, ELT(dbt), Data Lakes, Data Warehousing, Orchestration, Migration, Unit Testing

PUBLICATIONS: [Google Scholar](#)

- Location Prediction and Trajectory Optimization in Multi-UAV Application Missions **May 2022**
Book: Communications for Public Safety Networks | Chapter, pp105-131 | Publisher: Springer
- Environmentally Aware and Energy-Efficient Multi-Drone Coordination and Networking for Disaster Response **June 2023**
Journal: Transactions on Network and Service Management | Volume: 20, Issue: 2 | Page(s): 1093 - 1109 | Publisher: IEEE

PROJECTS:

Multi-Agent Drone Optimization using Deep Reinforcement Learning

Implemented Actor-Critic (A3C) models to optimize routing and decision-making for simulated real-world scenarios

Tech: Python, TensorFlow, OpenAI Gym, Numpy, Pandas

Link: <https://github.com/singh-rounak/Reinforcement-Learning-for-Drone-Trajectory>

CERTIFICATIONS:

Snowflake - Hands-on Essentials - Data Applications | [Link](https://www.credly.com/go/nco14Jqm) - <https://www.credly.com/go/nco14Jqm>

Snowflake - Hands-on Essentials - Data Warehouse | [Link](https://www.credly.com/go/cfJSh2E0) - <https://www.credly.com/go/cfJSh2E0>